# LabPath Practice Report

End-to-End Completion of the Published Problem Catalogue

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| Scope | 30 live problems — 18 RAG · 11 Guardrails · 1 Agent |
| Platform | LabPath (simulator + hidden eval graders) |

## Executive Summary

I completed the full published LabPath catalogue. Each exercise uses a production-style brief, an interactive simulator, structured submission, and automated graders with a hidden evaluation set. This report covers what the problems are like, how I executed the catalogue, challenges encountered, and outcomes.

## 1. What the problems are like

- **Brief** — Scenarios framed as launch reviews, security drills, or eval regressions.
- **Simulator** — RAG pipeline (corpus, chunk, query, generate), guardrails L1/L2/L3, or agent/tool loops.
- **Submission** — JSON fields: chunk params, prompts, attacks, page content, retriever code, controls.
- **Grading** — Class A gates: recall@k, nDCG, citations, refusals, levels won, Wilson thresholds, tool logs.
- **Feedback** — Sample failing cases on miss; hidden eval decides pass. No partial credit on class A.

Difficulty: Easy introduces one lever; Medium adds trade-offs; Hard mirrors production (hybrid retrieval, adversarial corpus, slice regression, multi-turn jailbreak, custom retriever code).

## 2. RAG track (18) — nature of the problems

| Phase | Focus | What they were like |
|---|---|---|
| 1 | Corpus | Chunk boundaries, overlap, top-k, scope — wrong splits hid answers from context. |
| 2 | SLA / cost | Policy KB recall; token and spend caps — recall-only tuning failed. |
| 3 | Generation | Frozen retrieval; [chunk:id] citations, grounding, refusals; stale index; hidden vs public eval. |
| 4–6 | Production | Rerank, hybrid, code retriever, packing, multi-hop, adversarial spans, slice regression gate. |

## 3. Guardrails track (11) — nature of the problems

| Mode | # | What they were like |
|---|---|---|
| G1 | 4 | Three-level concierge; attack prompts — direct exfil, indirect via retrieval, encoding, supply-chain docs. |
| G2 | 2 | Tool/data paths — BCC smuggle on send_email, PII/secret canaries via content the agent reads. |
| G3 | 5 | Blue-team — refusal UX, rules+judge, Wilson FP/FN, benign pass, multi-turn escalation. |

## 4. Agent track (1)

**Agentic Loop with Grader Stop** — Malformed tool calls require one recovery (corrected args) then stop; pass verified from tool logs (no killed_loop).

## 5. Method

- RAG phases in catalogue order; Guardrails G1→G2→G3; agent last.
- Submission log: change → metric → hidden pass/fail.
- Review class A gates before tuning.
- Time-boxed sessions on Hard problems.

## 6. Challenges

| Area | Issue |
|---|---|
| Hidden eval | Public pass, hidden fail until strategy changed. |
| R3 citations | False refusals on answerable questions. |
| Guard layers | L1 attacks failed L2/L3 without redesign. |
| Metrics | Recall tuning broke cost/spend problems. |
| G3 | Wilson / benign-pass threshold sensitivity. |
| Environment | Production VM auth/cookies cost session time (ops, not content). |

## 7. What I learned from solving the problems

7.1 RAG track

| Topic | Learning |
|---|---|
| Chunking | Wrong boundaries hide answers in context; heading/sentence splits beat fixed windows; overlap after sane splits. |
| Metrics & cost | recall@k is a contract; log k vs tokens; FinOps caps conflict with recall-only tuning. |
| Generation | Citations [chunk:id], grounding, and refusal balance — over-refusal fails answerable cases. |
| Eval | Public smoke ≠ ship; hidden eval and slice regression gates (aggregate up, slice down). |
| Advanced | Rerank/hybrid trade latency; custom retriever = explicit scoring; adversarial/multi-hop = real KB risk. |

7.2 Guardrails track

| Topic | Learning |
|---|---|
| G1 | Per-level attacks — L1/L2/L3 need different shapes; one jailbreak rarely clears all. |
| G2 | Risk in page/tool context; BCC smuggle & canaries = schema validation, not chat-only. |
| G3 | Rules+judge, Wilson FP/FN, benign pass — tune monitors like production alerting. |
| Mindset | G1/G2 offence informs G3 control tuning — need real attack shapes, not guesswork. |

7.3 Agent & process

- **Agent:** Recover once with different tool args, then stop; grades from tool logs, not prose.
- **Process:** Class A gates are the spec; failing-case snippets beat generic tips.
- **Habit:** Submission log (change → metric → pass/fail) on Hard problems.
- **Reviews:** Full catalogue = map of skills (chunking through Wilson gates) with concrete examples.

## 8. Conclusion

I completed all 30 published LabPath problems. The catalogue covers retrieval engineering, generation policy, eval discipline, offensive and defensive LLM security, and agent loop recovery. The learnings above are what I would apply on the next production RAG or agent launch.